

Q1.

For this project, I selected Cryptochrome-1 (CRY1) from Homo sapiens as my starting protein.

The RefSeq accession number for this protein is NP_004066.1, or Q16526 for UniProt.

Q2.

Among the top hits, a strong match was identified as **Bos taurus cDNA clone**

LB01721.CR_O20 GC_BGC-17 IMAGE:8095822 5', mRNA sequence (GenBank accession DV817148.1). The sequence was annotated only as a cDNA clone with no defined functional name, making it a suitable candidate for a potentially novel homolog of human CRY1.

The alignment between the human CRY1 query and the Bos taurus sequence showed **an E-value of 0.0, a bit score of 609, and 99.3% sequence identity across a 901-nt region**. These results indicate a very high level of homology, suggesting that this Bos taurus clone encodes a protein closely related to human Cryptochrome-1.

LB01721.CR_O20 GC_BGC-17 Bos taurus cDNA clone IMAGE:8095822 5', mRNA sequence

Sequence ID: [DV817148.1](#) Length: 901 Number of Matches: 1

Range 1: 8 to 901 [GenBank](#) [Graphics](#)

[▼ Next Match](#) [▲ Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
609 bits(1571)	0.0	Compositional matrix adjust.	296/298(99%)	297/298(99%)	0/298(0%)	+2
Query 1		MGVNAVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINRWRFLQCLE				60
Sbjct 8		MGVN+VHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGINRWRFLQCLE				187
Query 61		DLNANLRKLNLSRLFVIRGQPADVPRLFKERNITKLSIEYDSEPFGERDAAIKKLATEA				120
Sbjct 188		DLNANLRKLNLSRLFVIRGQPADVPRLFKERNITKLSIEYDSEPFGERDAAIKKLATEA				367
Query 121		GVEVIVRISHTLYDLDKIIEELNGGQPPLTYKRFQTLISKMEPLEIPVETITSEVIEKCTT				180
Sbjct 368		GVEVIVRISHTLYDLDKIIEELNGGQPPLTYKRFQTLISKMEPLEIPVETITSEVIEKCTT				547
Query 181		PLSDDHDEKYGVPSLEELGFDTDGLSSAVWPGGETEALTRERHLERKAWVANFERPRMN				240
Sbjct 548		PLSDDHDEKYGVPSLEELGFDTDGL SAVWPGGETEALTRERHLERKAWVANFERPRMN				727
Query 241		ANSLASPTGLSPYLRFGCLSCRLFYFKLTDLYKKVKNSSPPLSLYGQLLWREFFYT				298
Sbjct 728		ANSLASPTGLSPYLRFGCLSCRLFYFKLTDLYKKVKNSSPPLSLYGQLLWREFFYT				901

Q3.

Protein sequence:

>putative_CRY1_like|Bos_taurus|DV817148.1|ORF2|frame+2|len=297aa

MGVNSVHWFRKGLRLHDNPALKECIQGADTIRCVYILDPWFAGSSNVGIN

RWRFLQCLEDLDNANLRKLNLSRLFVIRGQPADVPRLFKERNITKLSIEY

DSEPFGERDAAIKKLATEAGVEVIVRISHTLYDLDKIIEELNGGQPPLTY

KRFQTLISKMEPLEIPVETITSEVIEKCTTPLSDDHDEKYGVPSLEELGF

DTDGLPSAVWPGGETEALTRLERHLERKAWVANFERPRMNANSLASPTG

LSPYLRFGLSCRLFYFKLTDLYKKVKKNSSPPLSLYGQLLWREFFYT

Name (provisional): Putative cryptochrome-1-like protein

Species: *Bos taurus* (from cDNA clone LB01721.CR_O20 GC_BGC-17 IMAGE:8095822; GenBank DV817148.1)

How obtained: The nucleotide sequence DV817148.1 was translated with NCBI ORFfinder (standard code; ATG start; six-frame translation). I selected the longest ORF (ORF2, + strand, frame +2, 894 nt / 297 aa) with no internal stops.

Q4.

As shown by figure below, none of my hits had 100% identity to this protein. The best BLASTP hit is Cryptochrome-1 from *Bos mutus* (ELR57829.1) with 99.66% identity, 0.0 E-value, and 0 gaps over 298 aa. Because the top match is <100% identical and from a different species than the source clone (*Bos taurus*), this satisfies the project's definition of a novel homolog.

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Cryptochrome-1 [Bos mutus]	Bos mutus	615	615	100%	0.0	99.66%	589	ELR57829.1
☒	cryptochrome-1 isoform X2 [Callorhinus ursinus]	Callorhinus ursinus	614	614	100%	0.0	99.66%	611	XP_073756447.1
☒	cryptochrome-1 isoform X2 [Pteronotus mesoamericanus]	Pteronotus mesoamericanus	614	614	100%	0.0	99.66%	616	XP_054430553.1
☒	cryptochrome-1 isoform X4 [Vicugna pacos]	Vicugna pacos	614	614	100%	0.0	99.66%	627	XP_072829164.1
☒	cryptochrome-1 isoform X2 [Odocoileus virginianus]	Odocoileus virginianus	614	614	100%	0.0	99.66%	628	XP_070310107.1
☒	cryptochrome-1 isoform X1 [Capricornis sumatraensis]	Capricornis sumatraensis	614	614	100%	0.0	99.66%	628	XP_068826436.1
☒	cryptochrome-1 isoform X2 [Bubalus kerabau]	Bubalus kerabau	614	614	100%	0.0	99.66%	628	XP_055430804.1
☒	cryptochrome-1 isoform X2 [Bos taurus]	Bos taurus	614	614	100%	0.0	99.66%	628	XP_059742545.1
☒	cryptochrome-1 isoform X1 [Vicugna pacos]	Vicugna pacos	613	613	100%	0.0	99.66%	587	XP_006205926.1
☒	cryptochrome-1 isoform X1 [Perognathus longimembris pacificus]	Perognathus longimembris p...	613	613	100%	0.0	99.66%	587	XP_048212448.1
☒	cryptochrome-1 [Budorcas taxicolor]	Budorcas taxicolor	613	613	100%	0.0	99.66%	587	XP_052495686.1
☒	cryptochrome-1 isoform X2 [Capricornis sumatraensis]	Capricornis sumatraensis	613	613	100%	0.0	99.66%	627	XP_068826437.1
☒	cryptochrome-1 isoform X2 [Desmodus rotundus]	Desmodus rotundus	613	613	100%	0.0	99.66%	611	XP_053777980.1
☒	cryptochrome-1 isoform X2 [Dama dama]	Dama dama	613	613	100%	0.0	99.66%	628	XP_060981421.1